

CHAPTER 6

Optical Music Recognition

Now that we have created a dataset for *Baishidaoren Gequ*, we can turn our attention to the development of algorithms and techniques belonging to the realm of OMR. Accordingly, in this section, we are going to consider Research Question (RQ3): How can accurate algorithms for the OMR of *suzipu* and *lülüpu* notations be developed?

The OMR task for *suzipu* is challenging, since each notation symbol is made up of a pitch (eleven possibilities) and an optional secondary component (either none or one out of six), therefore yielding 77 classes in total. However, only 66 actually occur in the dataset, and their distribution is highly imbalanced as seen in Chapter 5. In addition, the appearance of symbols in some classes may vary considerably up to ambiguity. First, we develop models for the classification of *suzipu* symbols. In order to assess the model's performance, a user study is conducted, in which 15 human participants are asked to solve the symbol recognition task.

In comparison with *suzipu*, the *lülüpu* notation is simpler, since all 17 classes occur in the dataset. Since the notations consist of standard Chinese characters with a very limited amount of classes, OMR for this notation can be considered as fairly straightforward. To justify the need for a separate classifier even though the *lülüpu* notation consists of a subset of Chinese characters, the model performance is compared with out-of-the-box OCR methods.

For the most complex notation, *jianzipu*, the exact number of possible classes cannot be determined easily, but the 708 instances present in the dataset are made up of 98 different annotations across all three notation classes. Due to its complexity and the lack of exhaustive datasets, OMR of this notation is quite difficult, and the 708 instances that are found in *Baishidaoren Gequ* are too few to create OMR algorithms that perform well on *jianzipu*.

Therefore, we discuss the task of symbol classification and clustering for both the *suzipu* and *lülüpu* notation symbols based on image data. A detailed description of their challenges and solutions are found in this section. The source code is open-source and publicly available.¹

In our case, by symbol classification we mean the process of predicting the correct symbol ground truth given its bounding box and the corresponding image data. For this, using the dataset extractor script,² the image patches are extracted and stored as individual images with a JSON file describing their annotations.

¹<https://github.com/SuziAI/SuziOMR>

²The dataset extraction procedure is described in https://github.com/SuziAI/gui-tools/blob/v2.1/readme_files/README_ANNOTATION_TOOL.md#extract-omr-dataset-from-corpus.

6.1 Notation Clustering

Inside the *Chinese Musical Notation Annotation Tool*'s intelligent display widgets, an intelligent selection of already annotated KuiSCIMA dataset instances with the same annotation as the currently selected notation instance is implemented. The source code for this section is found online.³

In the case that an annotation occurs up to 39 times in the dataset, the displayed selection consists of all such instances. However, if there are more than 39 instances with this annotation in the dataset, the data is clustered into 39 classes using the K-means clustering algorithm. From each of those clusters, the first element is chosen as a representative.

The images for each annotation class are pre-generated and loaded into the GUI at startup to reduce loading times when using the tool. In addition, since this method expects only an annotation and requires no image data as input, it is implemented not only for the *suzipu* and *lülüpu* notations, but also for *jianzipu*.

An example of the exported selection embedded into the tool's intelligent assistant is seen in Figure 4.5, widget (5).

6.2 Model Architecture and Evaluation Methodology

Since both the *suzipu* and the *lülüpu* datasets are fairly small, a lightweight CNN is chosen to prevent overfitting. This has the additional benefit that inference times are very small, allowing the model to smoothly run on the CPU without the need of having dedicated GPUs available.

The model consists of three convolutional layers and two fully connected linear layers, taking inputs of 48×48 pixels. The convolutional layers are padded to preserve output dimensions, and after each of them, a ReLU (*rectified linear unit*) activation and a 2D batch normalization are applied. Then, a 2×2 max-pooling layer is employed to reduce the dimensionality of the input. The first fully connected layer `fc1` features a ReLU activation and a 50% dropout. Depending on the feature extractor, the output dimension is either 11 (pitch features) or 7 (secondary features). The network's structure is visualized in 6.1.

In the scenario we want to model, we are already given a set of already annotated editions, and the model's task is to learn how to classify notation symbols of another, unseen edition. Since each edition is typically handwritten (it is either a manuscript or a xylographic reproduction), the individual features of the instances may differ significantly as opposed to typeset documents, where a limited number of typefaces are used typically.

To prove the setup's suitability for this specific use case, a leave-one-out cross-validation approach is applied. For this, each of the five editions in the KuiSCIMA dataset is used as a test set. The training and validation sets are then formed by dividing samples according to their annotation (consisting of both pitch and secondary labels), where 75% of samples belonging to each annotation are randomly assigned to the training set and 25% to the validation set. With

³https://github.com/SuziAI/SuziOMR/blob/v2.0/baseline/00_notation_clustering.ipynb

⁴The visualization was created using Haris Iqbal's PlotNeuralNet (<https://github.com/HarisIqbal88/PlotNeuralNet>).

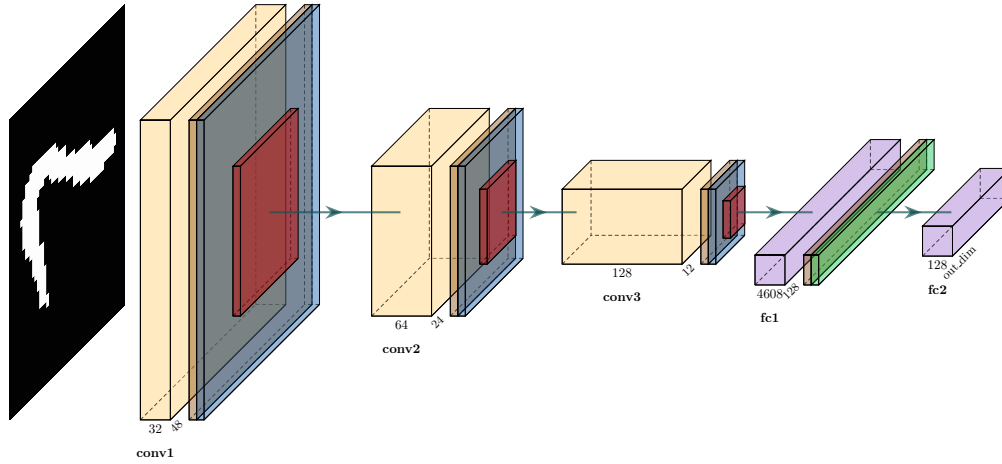


Figure 6.1: The small CNN features three convolutional layers and two fully connected layers. Convolutions are yellow, ReLU activations are orange, batch normalization is blue, max-pooling is red, fully connected layers are lavender, and dropout is green.⁴

this, the training and validation sets are guaranteed to have similar compositions with respect to their ground truth classes.

To identify each of those five model groups, we refer to them by the edition that constitutes their test set. E.g., the `Lu` model is tested on the `Lu` edition instances, and trained (respectively validated) on every other edition. Similarly, the `Zhang`, `Siku`, `Zhu`, and `Shanghai` models are tested on the `Zhang` edition, the `Siku Quanshu` edition, the `Zhu` edition, and the `Shanghai MS` respectively.

Since training a model is a stochastic process (depending on random parameters such as model initialization, training-validation split, batch composition, etc.), each trained model can be seen as a collection of random variables constituting the model weights. Therefore, ten distinct models are trained for each cross-validation group and each feature type (i.e., either pitch or secondary features). In total, this yields $10 \times 2 = 20$ model samples.

6.3 Classifier for *Suzipu*

The KuiSCIMA dataset contains 7297 non-empty annotated *suzipu* notation instances, from which 133 are excluded from training due to anomalous shapes, as described in Section 5.4. Of these, 5168 are simple symbols (meaning that no secondary property is present), and 1996 are composite symbols. The source code for this section is found online.⁵

⁵https://github.com/SuziAI/SuziOMR/blob/v2.0/baseline/01_suzipu_models.ipynb

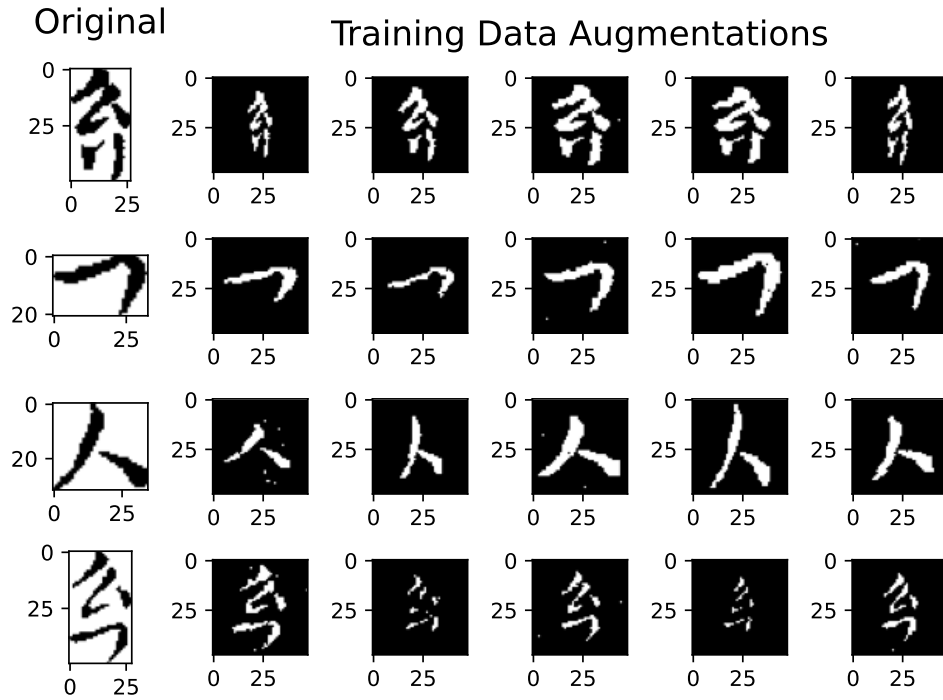


Figure 6.2: These data augmentations are used for training the model. The first column contains the image as directly exported from the dataset, and the other columns contain some random samples of augmentations of the images in the first column.

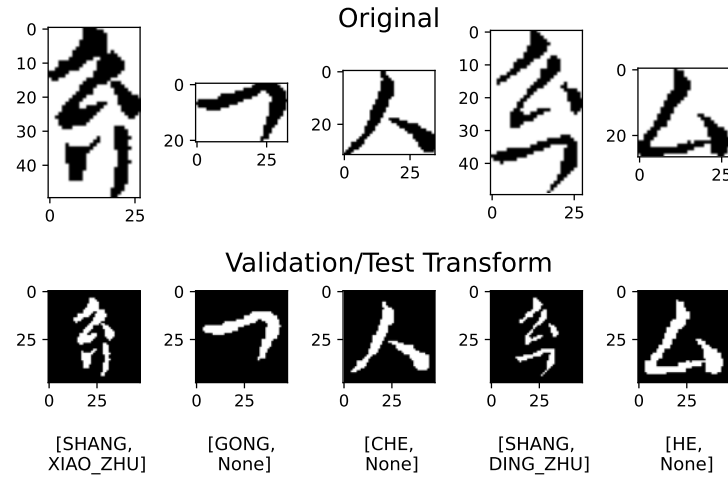


Figure 6.3: Here, the transformation of the validation and test data is seen. The first row contains the image as exported from the dataset, the second row contains the corresponding transformation, and the third row contains the ground truth annotation.

6.3.1 Image Preprocessing

Regarding preprocessing, each of the image instances is first cleaned up by removing small connected areas which might be printing artifacts. For this, we extract the contours using OpenCV's (Bradski, 2000) `findContours` function with attributes `cv2.RETR_EXTERNAL` and `cv2.CHAIN_APPROX_NONE`. Areas which have a `contourArea` smaller than 1 are removed. Afterwards, the image is cropped such that white space on the borders is removed.

For training images, additional image augmentations are used. To account for difference in stroke thickness, erosion and dilution with a 2×2 kernel are applied with a probability of 20% each. Then, the image is shrunk such that the longest side is uniformly distributed between 30 and 42 pixels, and the aspect ratio is multiplied with a uniformly distributed factor between 0.6 and 1.5. The image is then evenly extended by white space to fit into a 48×48 square, salt-and-pepper noise is applied, it is inverted to have black background and white foreground (this is important since the model contains max-pooling layers), random rotation between -9 and 9 degrees is applied, and the image is normalized with the dataset mean of 0.7102 and standard deviation of 0.0914. The training data augmentations are visualized in Figure 6.2. The validation and test images are shrunk such that the longest side has 40 pixels, extended to a square and normalized as described before. This can be seen in Figure 6.3.

6.3.2 Model Training

Each of the pitch and secondary property are predicted using separate instances of the model, with 11 output parameters for pitch, and 7 for secondary (since a secondary annotation can either be one out of the six secondary symbols or their absence). As such, the classifier consists of two small CNNs leveraging this special structure of the *suzipu* notation. The first classifier is for the pitch component, while the second classifier deals with the notation's secondary component. Similar methods have already successfully been used in other settings where the classes can be visualized as a product space, e.g. the Ethiopic script as demonstrated by Belay et al. (2019). For each of those components and for each of the test editions, 10 models are trained.

Each model sample is trained with 80 epochs of 43 batches with 100 images each. In order for the classifier to learn features of all classes, the training instances are sampled uniformly (with replacement, but with the aforementioned data augmentations) with respect to the feature class the classifier is trained on. I.e., for a pitch classifier, the training instances are sampled such that the distribution of the pitch labels is uniform. The validation and test instances are traversed completely and without replacement.

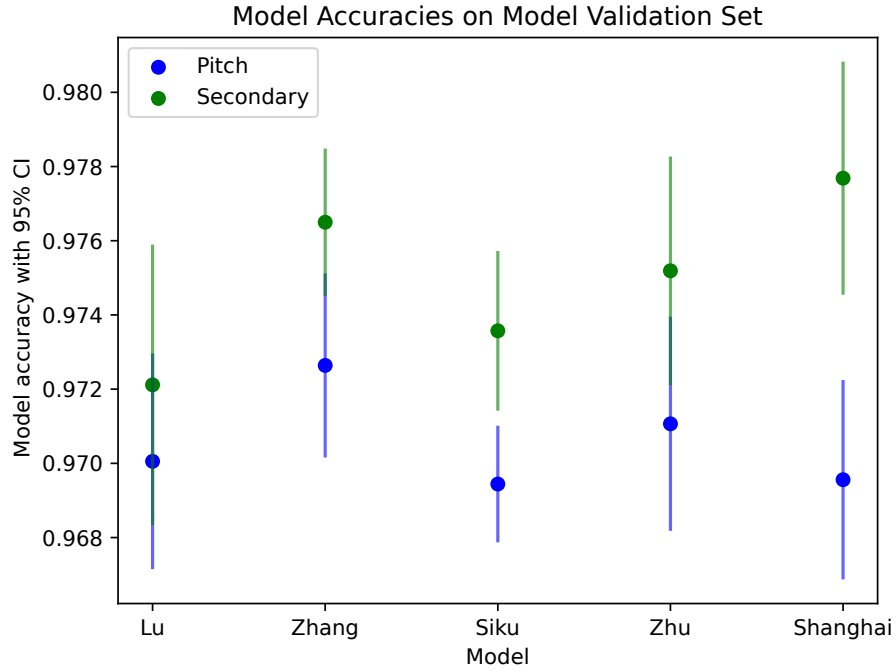
The optimizer *Adam* with a learning rate of 10^{-3} is used. In addition, a weight decay of 10^{-4} is used on all of the model's parameters except biases and batch normalization. In case the validation loss plateaus for five epochs, the learning rate is reduced by a factor of 0.5. For the loss function, focal loss by Lin et al. (2020) is chosen with $\gamma = 1$, since it reduces the loss of well-classified examples and therefore allows the model to learn from a small number of hard samples effectively, which is beneficial in settings of class imbalance.

Test Edition	Validation Acc. (%)		Test Acc. (%)			Test CER (%)
	Pitch	Secondary	Pitch	Secondary	Total	Total
Lu	97.0 $\pm$ 0.5	97.2 $\pm$ 0.6	96.5 $\pm$ 0.4	97.8 $\pm$ 0.3	94.5 $\pm$ 0.4	5.5 $\pm$ 0.4
Zhang	97.3 $\pm$ 0.4	97.7 $\pm$ 0.3	96.5 $\pm$ 0.7	96.8 $\pm$ 0.3	94.3 $\pm$ 0.7	5.7 $\pm$ 0.7
<i>Siku Quanshu</i>	96.9 $\pm$ 0.2	97.4 $\pm$ 0.3	96.4 $\pm$ 0.4	96.8 $\pm$ 0.5	93.9 $\pm$ 0.6	6.1 $\pm$ 0.6
Zhu	97.1 $\pm$ 0.5	97.5 $\pm$ 0.5	94.4 $\pm$ 0.6	96.0 $\pm$ 0.4	91.6 $\pm$ 0.6	8.4 $\pm$ 0.6
Shanghai MS	97.0 $\pm$ 0.4	97.8 $\pm$ 0.5	95.8 $\pm$ 0.5	96.1 $\pm$ 0.5	92.6 $\pm$ 0.6	7.4 $\pm$ 0.6
Aggregated			95.9 $\pm$ 1.0	96.7 $\pm$ 0.7	93.4 $\pm$ 1.2	6.6 $\pm$ 1.2

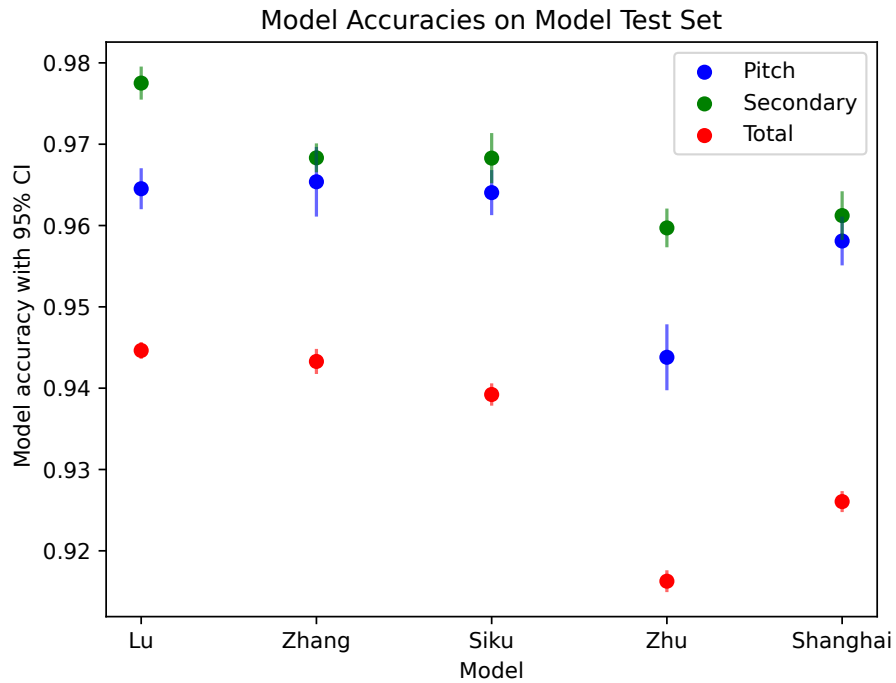
Table 6.1: For the *suzipu* classifiers, the average validation and test accuracies with standard deviations are shown in the table. Note that the number of samples is 10 for the individual classifiers, and 100 for the total classifiers. The test CER values are highlighted in boldface.

Test Edition	Validation Acc. (%)		Test Acc. (%)			Test CER (%)
	Pitch	Secondary	Pitch	Secondary	Total	Total
Lu	97.7	98.2	97.0	97.3	94.6	5.4
Zhang	97.8	98.4	97.5	97.3	95.4	4.6
<i>Siku Quanshu</i>	97.3	97.9	96.5	97.7	94.7	5.3
Zhu	97.7	98.3	93.6	95.9	91.0	9.0
Shanghai MS	97.8	98.8	95.8	96.6	92.9	7.1

Table 6.2: For the *suzipu* classifiers, the validation accuracies, test accuracies, and test character error rate (CER) are displayed for the best models (selected according to the validation accuracy). Note that the number of samples is 10 for the individual classifiers, and 100 for the total classifiers. The test CER results are printed in boldface.



(a) Validation set accuracies



(b) Test set accuracies. Note that there are 10 samples each for the pitch and secondary values, and 100 samples for the total models, which are created by combining a pitch and a secondary model.

Figure 6.4: *Suzipu* validation and test set accuracies for each model class with their means and 95%-confidence intervals

6.3.3 Results

In Table 6.1, the average validation and test accuracies are shown for each of the individual pitch and secondary classifiers (10 samples each per model class), as well as the total classifiers (100 samples per model class, since each total model consists of a pitch classifier and a secondary classifier). The aggregated total CER is $6.6\% \pm 1.2\%$. Additionally, Figure 6.4 shows the validation accuracies for the individual pitch and secondary classifiers, and the test accuracies for the total classification task.

As often done in practice, for each of the individual classifiers for each test set, the best performing model on the validation set is selected and then evaluated on the test set. These results are shown in Table 6.2, and it can be seen that the total CER for the *suzipu* OMR task is between 4.6% and 9.0%, depending on the test set.

The lowest value is achieved for the Zhu edition as test set for the averaged values. In addition, also when selecting the best model according to validation accuracy, the resulting model is worse than the average for this edition. This indicates that the Zhu edition has unique features that cannot be properly approximated by the other editions.

For timing, an instance of the total model (i.e., both the pitch and secondary classifiers) is 500 times on the Shanghai MS (which contains 1439 instances). The tests are conducted using an AMD Ryzen 7 PRO 4750U processor. This yields an average time of 2.04 ± 0.22 seconds for a single evaluation of the whole edition.

6.4 Classifier for *Lülüpu*

The *lülüpu* notation dataset used for training is introduced in KuiSCIMA v2.0 and contains 3385 non-empty *lülüpu* instances. The source code for this section is found online.⁶

6.4.1 Image Preprocessing

The *lülüpu* notation uses Chinese characters for denoting pitches, which results in more fine-grained image features. Therefore, no denoising is applied to the image patches. The other steps remain the same except for the image dimensions in the individual steps to ensure a better capture of the fine character details. This means that for the shrinking step, the longest side is uniformly distributed between 33 and 46 for the training images, and 40 for the test images. The final image size for model input is again 48×48 pixels.

For training images, the augmentations from the *suzipu* classifier are applied. Finally, the image is normalized with the dataset mean of 0.7148 and a standard deviation of 0.0575. Here, the training data augmentations are visualized in Figure 6.5.

To counteract the small dataset size, two classes of models are trained: One with original data only, and another one with artificially generated training data. For four different computer fonts (*AR PL UKai CN*, *AR PL Mingti2L Big5*, *Noto Sans CJK JP*, *Noto Serif CJK JP*) each, all 17 classes are generated with a font size of 60 and then preprocessed the same way as the original images. These fonts are chosen to expose the model to a variety of Chinese

⁶https://github.com/SuziAI/SuziOMR/blob/v2.0/baseline/01_lvlpvu_models.ipynb

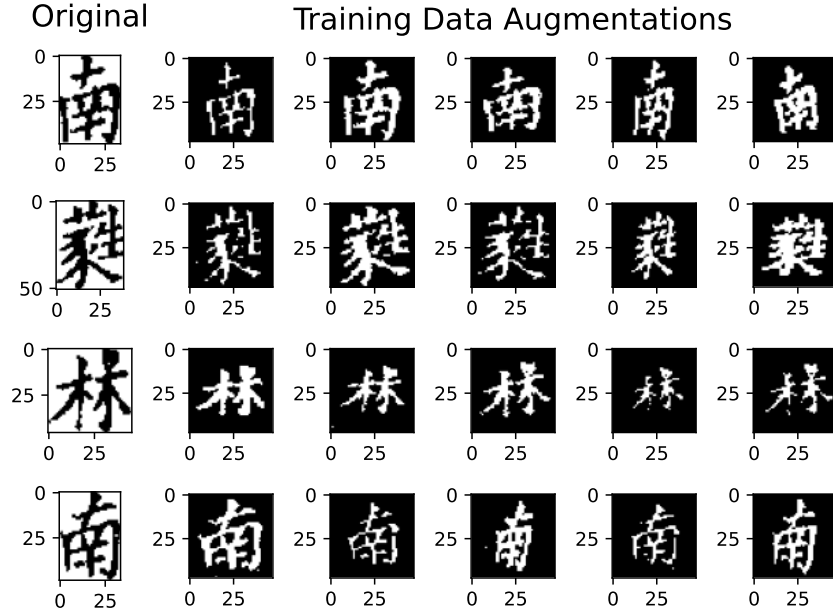


Figure 6.5: The first column contains the image as exported from the dataset, and the other columns contain some examples of image augmentations.

character styles, with the intention of reducing overfitting. In addition, whenever supported by the fonts, the two characters *huang* 黃 and *qing* 清 are additionally included with their variant characters *huang* 黃 and *qing* 清. This yields additional 86 training images, which are then included into the training dataset. For each of these four fonts, the first *lülüpu*'s symbol for *Huangzhong* is displayed with some data augmentations in Figure 6.6.

Again, the validation and test images are shrunk such that the longest side has 40 pixels, extended to a square and normalized with the dataset mean and standard deviation. The validation and test transformations are seen in Figure 6.7.

6.4.2 Model Training

Since the classification task involves 17 distinct classes, the output dimension of the model is set to 17 accordingly. Using 50 epochs of 21 batches (respectively 22 batches for the training data containing artificially generated samples) with 100 images each, each model class is trained using either the training set with or without artificial data. Similar to the *suzipu* classifier, the training instances are sampled uniformly (with respect to their annotation) with replacement. Again, the validation and test instances are sampled without replacement.

The optimizer *Adam* with a learning rate of $5 \cdot 10^{-4}$ and a weight decay of 10^{-4} is used on all of the model's parameters except biases and batch normalization. The validation loss learning rate scheduling and the loss function are the same as for the *suzipu* classifier in Section 6.3.

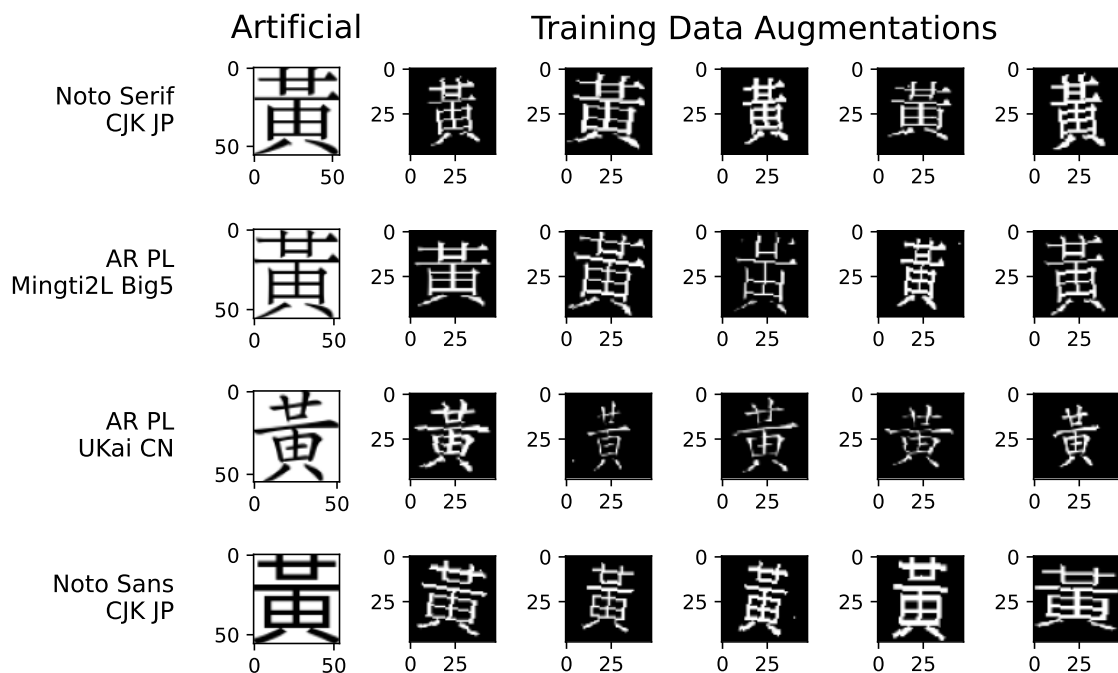


Figure 6.6: The font's name is in the first column, in the second column is the image generated from the font, and the other columns contain some training image augmentations of the corresponding images.

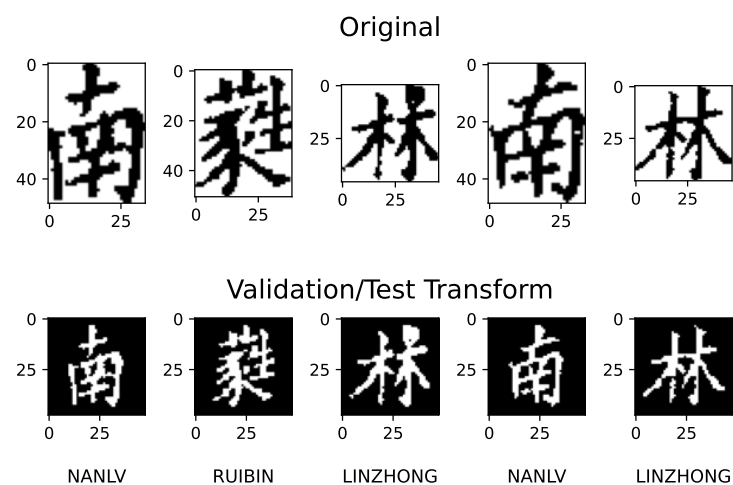


Figure 6.7: The first row contains the original image from the dataset, while the second row contains the corresponding deterministic transformation used for validation and testing. In the third row, the annotation is displayed.

Test Edition	Validation Acc. (%)		Test Acc. (%)		Test CER (%)	
	Non-Art.	Art.	Non-Art.	Art.	Non-Art.	Art.
Lu	99.4 $\pm$ 0.2	99.6 $\pm$ 0.2	99.5 $\pm$ 0.2	99.7 $\pm$ 0.2	0.5 $\pm$ 0.2	0.3 $\pm$ 0.2
Zhang	99.4 $\pm$ 0.2	99.5 $\pm$ 0.2	99.4 $\pm$ 0.2	99.7 $\pm$ 0.2	0.6 $\pm$ 0.2	0.3 $\pm$ 0.2
<i>Siku Quanshu</i>	99.4 $\pm$ 0.3	99.4 $\pm$ 0.2	99.3 $\pm$ 0.4	99.5 $\pm$ 0.3	0.7 $\pm$ 0.4	0.5 $\pm$ 0.3
Zhu	99.6 $\pm$ 0.3	99.6 $\pm$ 0.3	95.1 $\pm$ 1.9	97.8 $\pm$ 1.3	4.9 $\pm$ 1.9	2.2 $\pm$ 1.3
Shanghai MS	99.6 $\pm$ 0.2	99.6 $\pm$ 0.2	98.2 $\pm$ 0.6	98.7 $\pm$ 0.4	1.8 $\pm$ 0.6	1.3 $\pm$ 0.4
Aggregated			98.3 $\pm$ 1.9	99.1 $\pm$ 0.9	1.7 $\pm$ 1.9	0.9 $\pm$ 0.9

Table 6.3: For the *lülüpu* classifiers, the average validation and test accuracies with standard deviations are shown in the table for both the training sets with (Art.) and without (Non-Art.) artificial data. For each model class, there are 10 samples. The best test CER values are printed in boldface.

Test Edition	Validation Acc. (%)		Test Acc. (%)		Test CER (%)	
	Non-Art.	Art.	Non-Art.	Art.	Non-Art.	Art.
Lu	99.7	99.9	99.7	99.6	0.3	0.4
Zhang	99.7	99.7	99.4	99.7	0.6	0.3
<i>Siku Quanshu</i>	99.7	99.9	99.3	99.6	0.7	0.4
Zhu	100.0	99.9	95.0	95.4	5.0	4.6
Shanghai MS	100.0	99.9	97.6	99.1	2.4	0.9

Table 6.4: For the *lülüpu* classifiers, the best validation accuracies, test accuracies, and CER are displayed for the best models (selected according to the validation accuracy), for both the training sets with (Art.) and without (Non-Art.) artificial data. The best test CER values are printed in boldface.

6.4.3 Results

In Table 6.3, for each of the models with and without artificial data (10 samples each per model class), the average validation and test accuracies are shown. Figure 6.8 contains the validation and test accuracies for the classifiers.

For each of the classifiers and for each test set, the best performing model on the validation set is selected and then evaluated on the test set. From Table 6.4 one can see that the aggregated CER for the *lülüpu* OMR task is 1.7% for the models without artificial data, while it is 0.93% when including the artificial data in training. Also, the introduction of artificial samples reduced the variance of the accuracy values, making these models more robust.

Similar to the results of the *suzipu* classifier, when using the Zhu edition as a test set, the achieved error rates are significantly higher, and the best model according to validation accuracy performs worse than the average accuracy. However, for the aggregated CER values, the mean accuracy and also the variance could be notably improved by using artificial data,

furthermore indicating that the Zhu edition's image features strongly differ from the other four editions.

Regarding timing, an instance of the model is evaluated 500 times, yielding an average time of 0.53 ± 0.07 seconds for a single evaluation of the whole Shanghai MS (which contains 677 *lülüpu* instances).

6.4.4 Comparison with Tesseract OCR

It is important to remember that the *lülüpu* notation uses a subset of Chinese characters, so a comparison of the trained model against an out-of-the-box OCR algorithms is meaningful. For these tests, the source code is found online.⁷

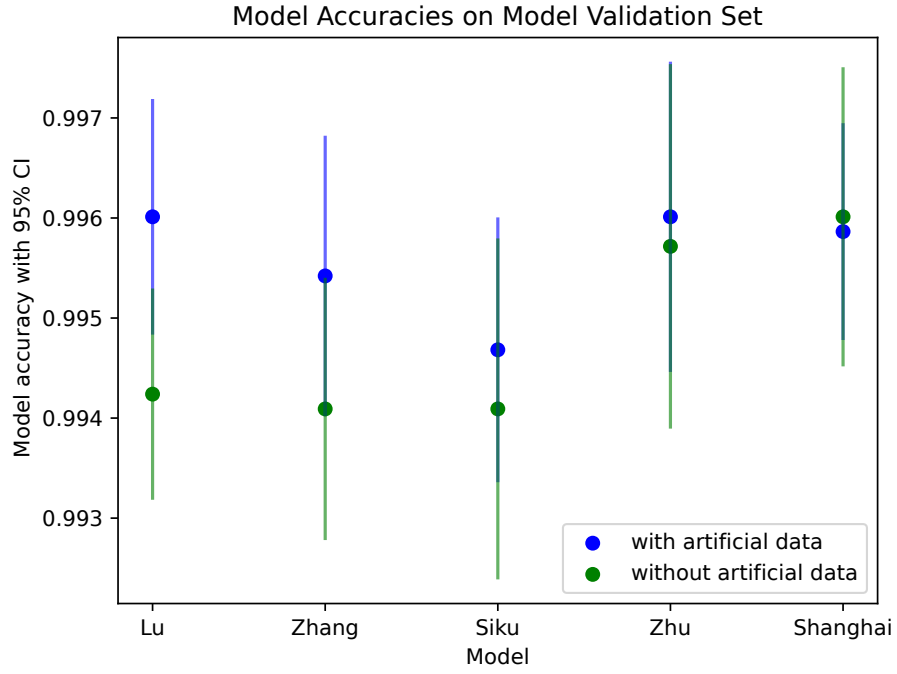
After preprocessing the images by resizing them to 60×60 pixels with cubic interpolation, applying a 3×3 gaussian blur, applying Otsu's method to obtain binary images, and padding the boundaries with 20 pixel whitespace on each of the four sides, the *Tesseract* OCR system (Kay, 2007) is used. The model file is `chi_tra.traineddata`⁸ file, which is the standard model for traditional Chinese text. The 15 distinct characters that appear in the *lülüpu* notation are whitelisted and tested with three different page segmentation modes (PSM). PSM 6 assumes a single uniform block of text, PSM 7 treats the image as a single text line, and PSM 8 treats the input as a single word. For all of the three PSM settings, spaces are removed from the OCR output, and the prediction is matched against the text of the *lülüpu* classes. Since the characters *zhe* 折 or *zi* 字 are both unique to the *zhezi* class, in case one of them appears in the Tesseract prediction, this class is returned. If there is no match between the model output and any *lülüpu* class, the most frequently appearing class *Linzhong* 林鐘, making up about 11.9% of the KuiSCIMA *lülüpu* dataset, is returned instead. The test is performed on the whole KuiSCIMA *lülüpu* dataset.

The CER using PSM 6 is 42.4%, for PSM 7 it is 42.1%, and for PSM 8 it is 44.0%. The timing test was performed with 15 iterations on the Shanghai MS dataset, yielding an average time of 7.9 ± 0.1 minutes.

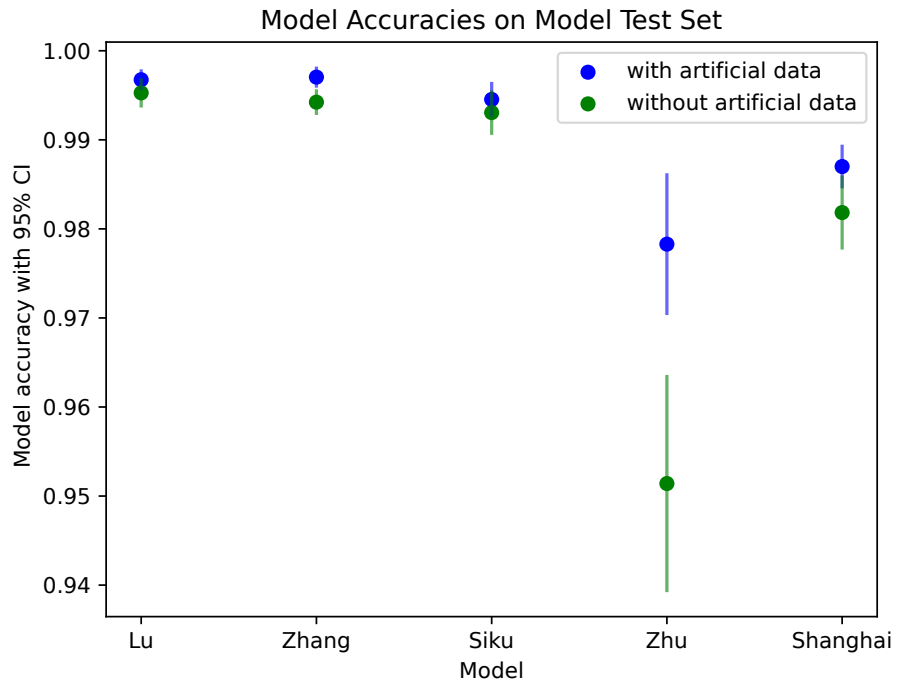
We assume that the reasons for these high CERs are threefold: Firstly, even though the Tesseract models are whitelisted, for Chinese character recognition a large number of characters must be supported, thus yielding lower performance on a small subset of characters. Secondly, the model was presumably trained on images of modern fonts predominantly, thus failing to deal with the handwritten historical texts. And thirdly, in textual sources the OCR can rely on context information to increase model performance, which is not applicable to the case of notation symbols.

⁷https://github.com/SuziAI/SuziOMR/blob/v2.0/baseline/04_lvlvpu_tesseract.ipynb

⁸https://github.com/tesseract-ocr/tessdata/blob/main/chi_tra.traineddata



(a) Validation set accuracies



(b) Test set accuracies. There are 10 samples each for the models with and without artificial data.

Figure 6.8: Lülüpu validation and test set accuracies for each model class with their means and 95% confidence intervals

Test Edition	<i>Suzipu</i> Pitch	<i>Suzipu</i> Secondary	<i>Lülüpu</i>	
	Uncalibr.	Uncalibr.	Uncalibr.	Calibr.
Lu	0.0101	0.0039	0.0038	0.0036 $\pm$ 0.0000
Zhang	0.0062	0.0039	0.0051	0.0049 $\pm$ 0.0000
<i>Siku Quanshu</i>	0.0089	0.0079	0.0167	0.0161 $\pm$ 0.0001
Zhu	0.0088	0.0154	0.0177	0.0162 $\pm$ 0.0001
Shanghai MS	0.0111	0.0139	0.0054	0.0044 $\pm$ 0.0001

Table 6.5: The aggregated ECE results for each of the models. The mean values of the calibrated are shown with their standard deviations, since ten calibrations per model were made. For the *lülüpu* classifier, the best values are highlighted in boldface.

6.5 Model Calibration

To make the OMR algorithm more interpretable to users of the GUI, the classifiers are calibrated using the temperature scaling approach with expected calibration error (ECE) by Guo et al. (2017) on their respective validation sets. The ECE is calculated on ten equidistant bins, and each calibration is done ten times to account for the stochastic nature of the training process. In addition, the source code is found on GitHub for both the *suzipu*⁹ and *lülüpu*¹⁰ models.

For the pitch classifier, the temperature is trained using the SGD optimizer with 5 or 10 epochs and a learning rate of 0.005 or 0.01. With all of those combinations, the ECE could not be reduced significantly when compared with the ECE of the uncalibrated models. Figure 6.9 visualizes how the models of the *suzipu* pitch and secondary classifiers are well-calibrated even without temperature scaling.

For the *lülüpu* classification model calibration, the SGD gradients vanished, probably due to the significantly smaller size of the dataset. Therefore, the Adam optimizer was used instead with 20 epochs and a learning rate of 0.001. A comparison of the reliability plots for the uncalibrated and calibrated models is seen in Figure 6.10.

The aggregated results for both the *suzipu* and *lülüpu* models are contained in Table 6.5.

6.6 Similarity Visualization

As an additional guidance for the user, the similarity visualization displays the three most similar notation instances found in the KuiSCIMA dataset with respect to their optical features. The visualizations can be generated with the source code on GitHub for both *suzipu*¹¹ and *lülüpu*¹² classifiers.

⁹https://github.com/SuziAI/SuziOMR/blob/v2.0/baseline/02_suzipu_temperature_scaling.ipynb

¹⁰https://github.com/SuziAI/SuziOMR/blob/v2.0/baseline/02_lvlvpu_temperature_scaling.ipynb

¹¹https://github.com/SuziAI/SuziOMR/blob/v2.0/baseline/03_suzipu_clustering_and_similarity.ipynb

¹²https://github.com/SuziAI/SuziOMR/blob/v2.0/baseline/03_lvlvpu_clustering_and_similarity.ipynb

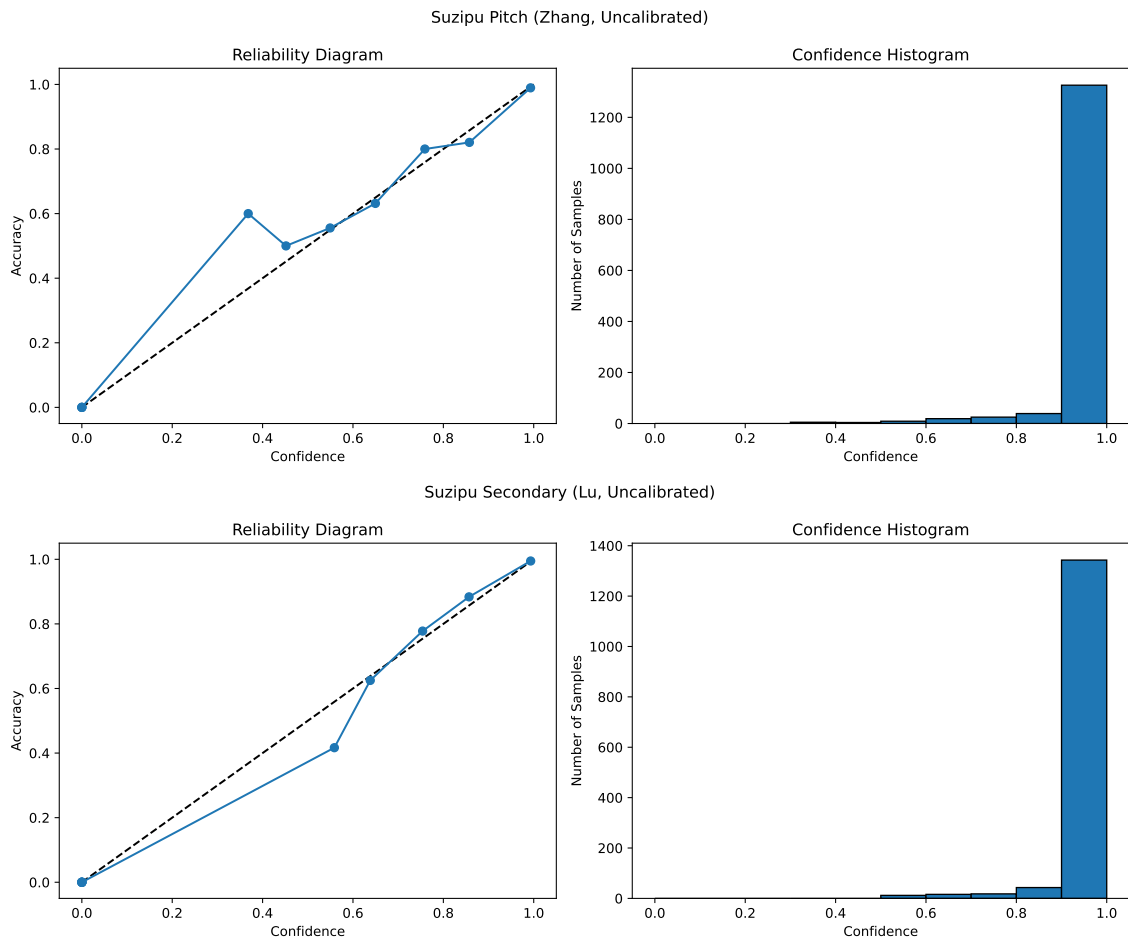


Figure 6.9: *Suzipu* OMR classifiers: The reliability plots indicate that the models are already well-calibrated: The pitch model has an ECE of 0.0062, while the secondary model has an ECE of 0.0039.

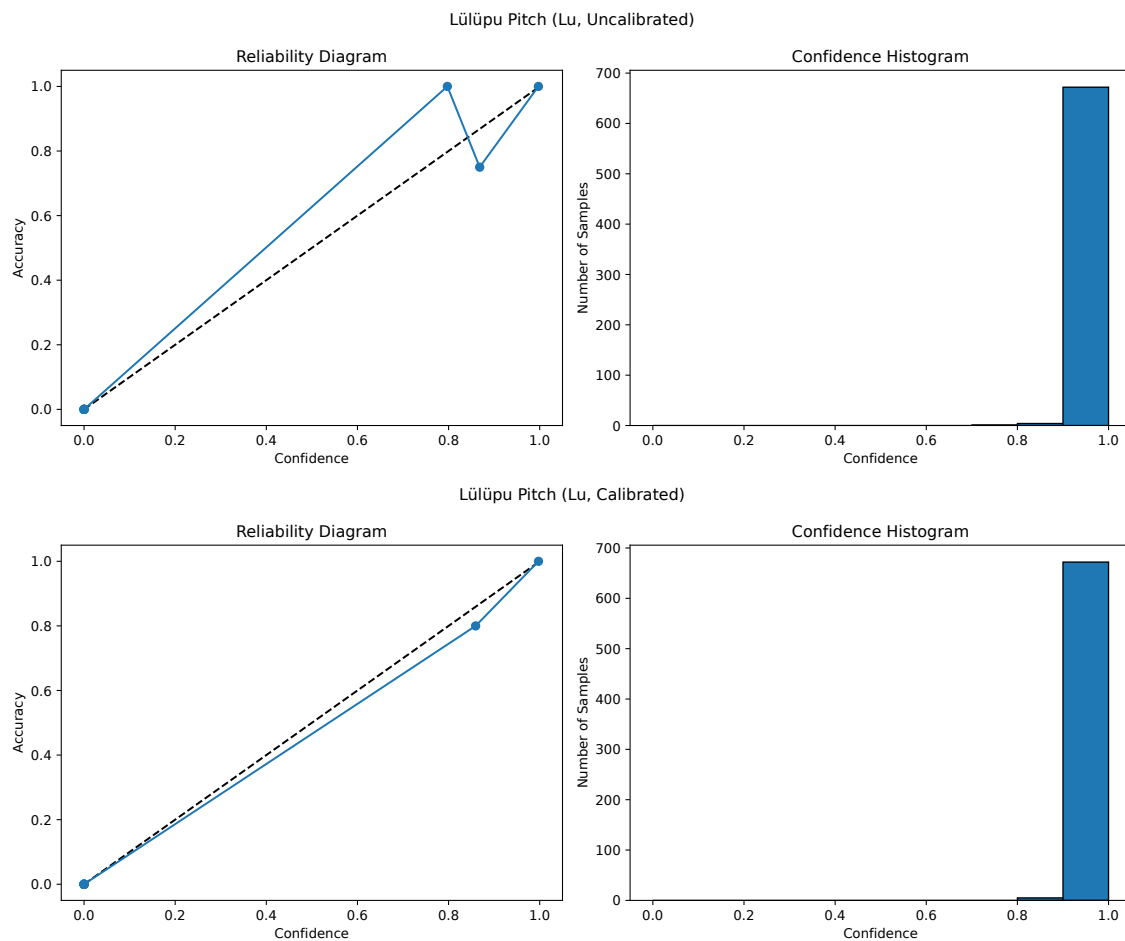
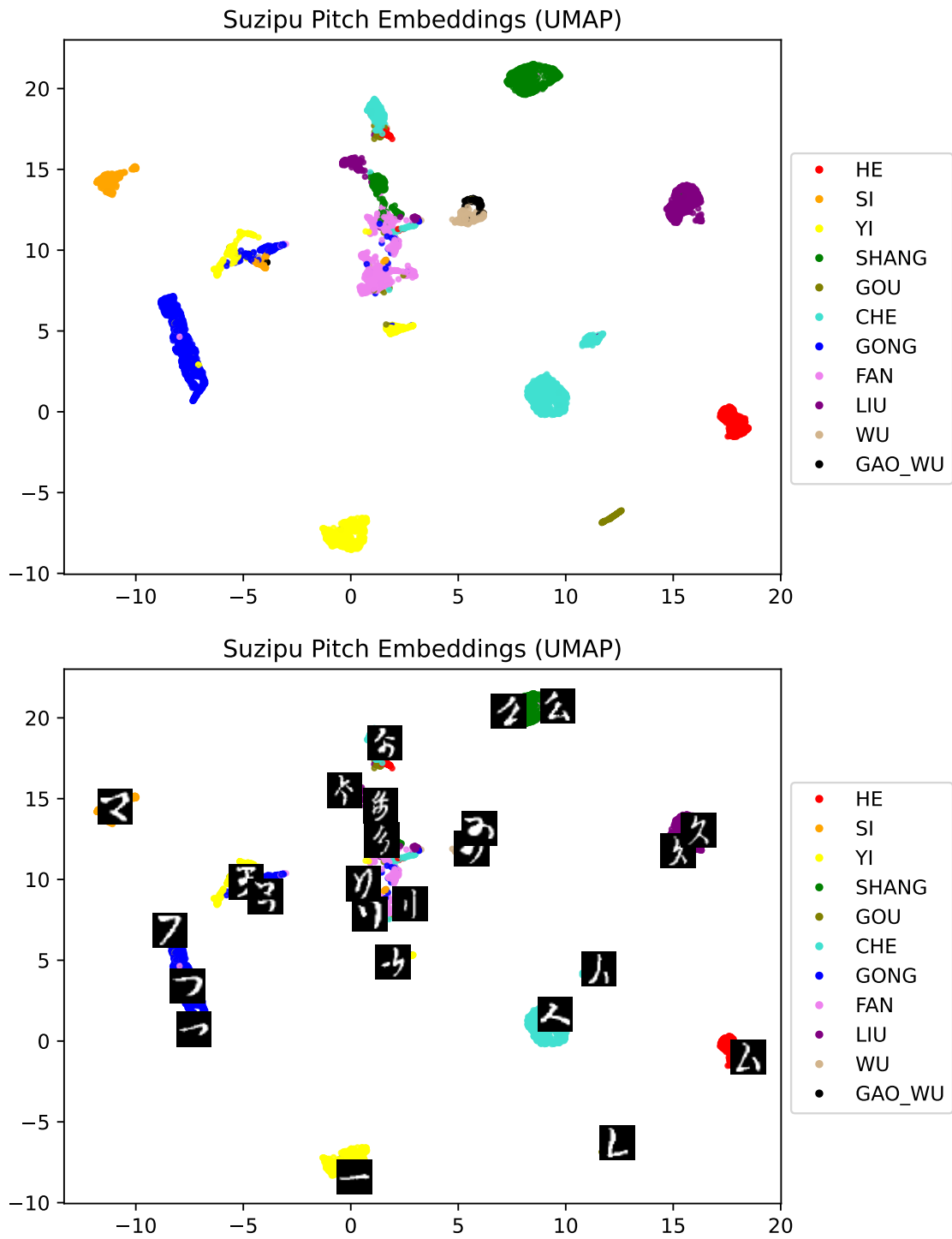


Figure 6.10: *Lülüpu* OMR classifier: The reliability plot on the Lu edition shows a good calibration. The uncalibrated ECE is 0.0038, while it is 0.0031 after temperature scaling with a temperature of 0.9765.

Figure 6.11: *Suzipu* pitch classifier UMAP embedding

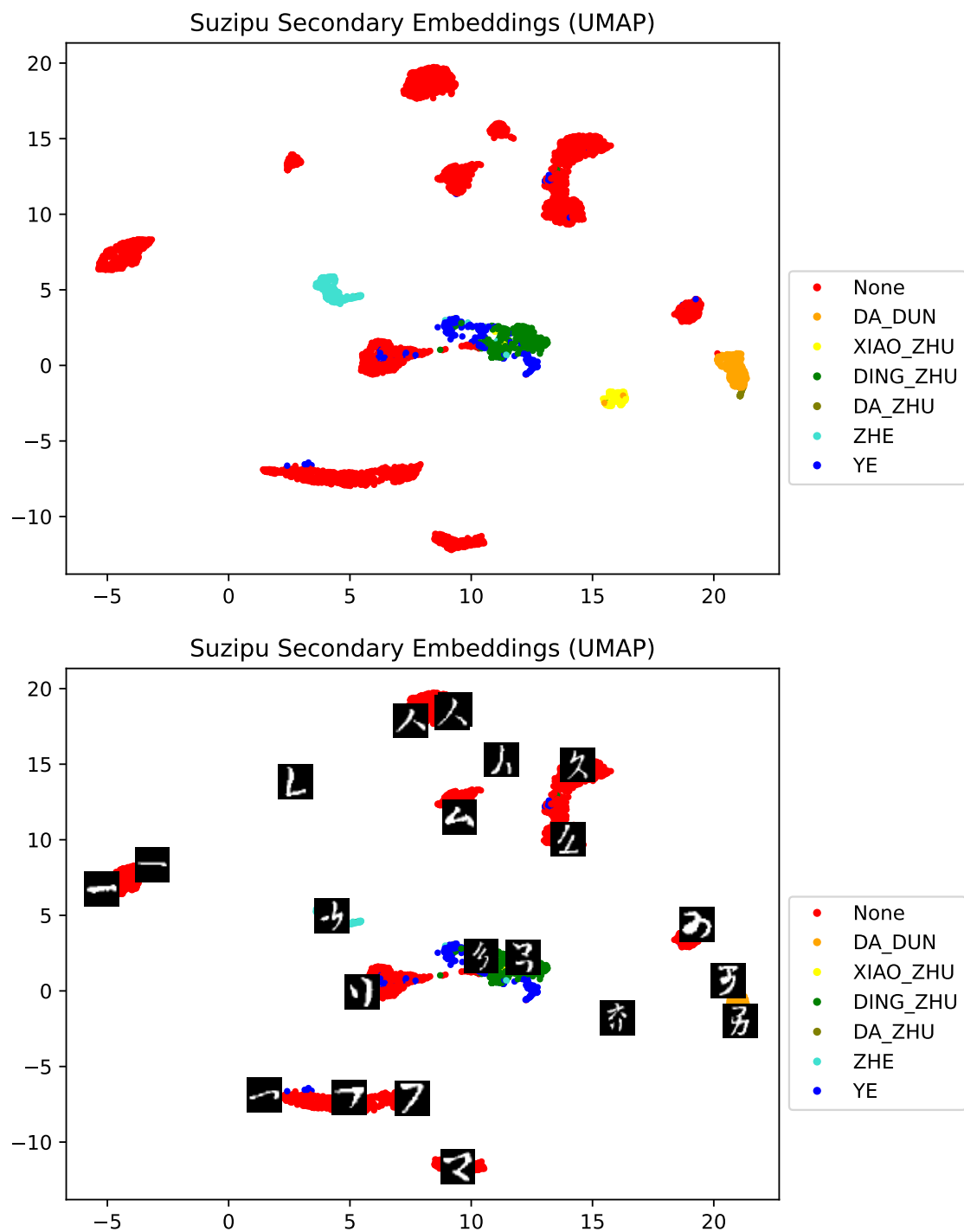


Figure 6.12: *Suzipu* secondary classifier UMAP embedding

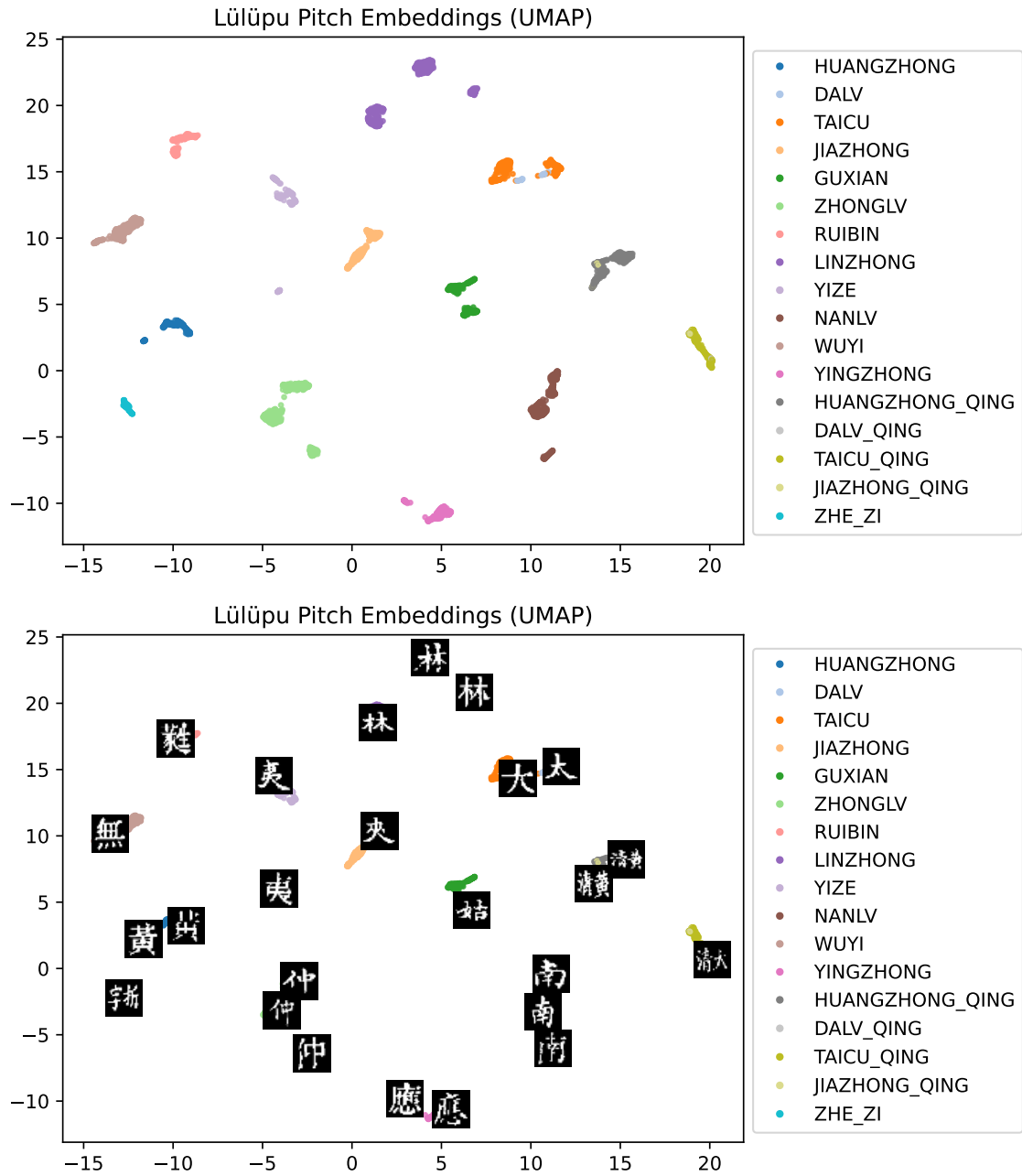


Figure 6.13: Lülüpu classifier UMAP embedding

In order to obtain a sharp separation between the individual clusters, the five best *suzipu* pitch and secondary models for each of the test editions obtained in Section 6.3, and the five best *lülüpu* models from Section 6.4 are applied to each individual sample of the KuiSCIMA dataset. Then, the best performing model is selected. This yielded the Zhang model (98.6% accuracy) for the *suzipu* pitch classifier, the Lu model (98.9% accuracy) for the *suzipu* secondary classifier, and the Lu model for the *lülüpu* classifier (99.82% accuracy).

For each of these three resulting OMR models, the layer `fc1` (which is a 128-dimensional vector) is extracted as a feature encoding of the respective property. The features are collected for each instance in the dataset, and an unsupervised UMAP (McInnes et al., 2018) dimensionality reduction to 2D is applied with a `random_state` of 42. Visualizations of the three UMAP spaces including samples from the dataset are found in Figures 6.11, 6.12, 6.13, where the coloring of the samples corresponds to the ground truth labels.

Then, the UMAP representation of the currently investigated notation instance is compared against the precomputed UMAP representation of all instances of the same notation type in the KuiSCIMA dataset, and the three nearest neighbors are retrieved, together with a similarity score that is calculated as the inverse euclidean distance between the current instance and the similar instance in the UMAP space.

An example of how the intelligent assistant shows these three nearest neighbors can be seen in Figure 4.5, widget (4).

6.7 *Suzipu* Classification Human-Level Performance

Due to the ambiguity of *suzipu* notation symbols, it is not immediately clear how well the model performs at the OMR task. Therefore, in order to assess the accuracy of the classifier created in this section, a user study is conducted, in which the participants are presented with the task of *suzipu* classification. With this, it is possible to compare the model’s performance with respect to classification accuracy and time with human performance.

Similar to the study for the Ethiopic script¹³ that was conducted by Belay et al. (2024), the human annotations are collected by the users at home using a specialized GUI which has been set up on a web server. The source code for the software used in this study is open-source and publicly available.¹⁴

Participants. A sample of 15 participants (7 female, 8 male; aged 33 ± 9.74 years) were recruited as volunteers through experiment postings. The participants have varying degrees of formal education. All of them have normal or corrected visual acuity. Participants were naive to *suzipu* notation and Chinese language, corresponding to settings as in the work of Belay et al. (2024), where in highly specialized tasks, a large number of minimally trained human annotators collect a dataset which is then corrected by a small number of highly skilled experts.

¹³https://github.com/bdu-birhanu/HHD-Ethiopic/blob/main/Supplementary_file/Revised_Historical_Handwritten_Ethiopic_dataset_Sup.pdf

¹⁴<https://github.com/SuziAI/HumanAnnotationBaselineStudy>

As compensation for the participants' time, a certificate and a bar of vegan chocolate have been provided.

Task description. The study consists of a collection of individual tasks, where each task corresponds to an image of an individual *suzipu* notation symbol. The goal for each of the tasks is to identify the correct pitch and secondary components in the *suzipu* framework as described in Section 3.3.

In Figure 6.14, the GUI with which the user interacts to solve the tasks is shown. The widgets are numbered and can be described as follows:

1. In this box, the notation symbol associated with the currently selected task is displayed.
- 2., 3. These two widgets are used for navigating the tasks. Clicking the button << Previous navigates to the previous task, while Next >> is used to get to the next task, according to their internal ordering.
4. This widget displays the index of the currently active task and the total number of tasks.
5. Using the pitch annotation widget, the user selects the *suzipu* pitch component of the image displayed in widget (1). In order for the current task to be completed, the pitch component selected must not be the leftmost box, which is empty.
6. Similar to widget (5), the user selects the *suzipu* secondary component associated with the displayed image. The leftmost button is empty to indicate the absence of a secondary component.
7. Clicking this button causes the user to be logged out of the session.
8. In this widget, the user's annotation progress is shown. If the displayed value reaches 100%, the measurement is successfully completed.
9. This widget is for fast navigation across all tasks. Red boxes indicate unseen tasks, yellow boxes indicate that the associated task has an unset pitch value in widget (5), and green boxes indicate completed tasks. By clicking a box, the user is taken to the corresponding task.

All of the tasks are taken from the Shanghai MS, which is the oldest preserved edition known to date, so the shape of the symbols is expected to be the clearest among all editions. Before the study, this edition's *suzipu* notations have been partitioned into two subsets according to their composite annotations (i.e., pitch and secondary), such that each of the two subsets have a similar distribution of samples according to their ground truth labels.

Each participant is given a sequential number starting from 0 according to the order of participation. Users with an even number are shown the samples from the first subset, while users with an odd number are shown the other samples. With this, the number of samples per user is not overwhelming while still guaranteeing a uniform coverage of all instances. The first subset contains 719 samples, while the second contains 720.

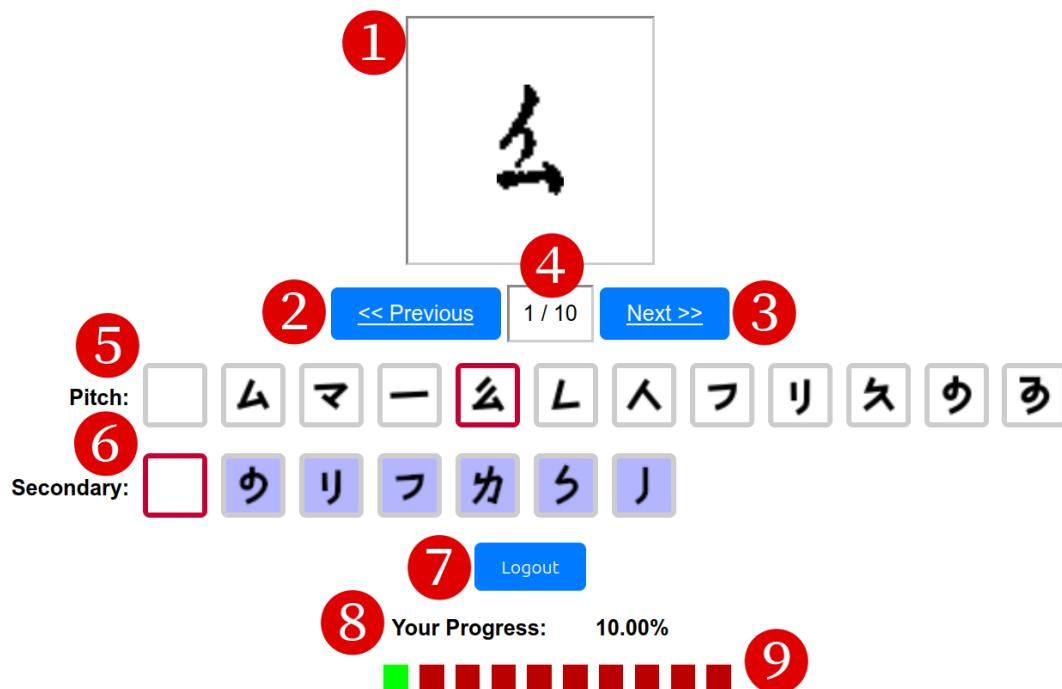


Figure 6.14: In this screenshot of the study GUI, the presented task is correctly solved. This image is taken from the training version of the GUI, which only differs from the measurement version by the amount of tasks.

Study design. In order to avoid travel costs and minimize the associated carbon footprint, the study is designed to be completed online on a participant's device with a sufficiently large screen (e.g., laptop, tablet computer or desktop PC) and internet connection.

The study itself consists of two phases: In the first phase (training phase), the user is asked to read the task description sheet (see Appendix E.1) and to familiarize themselves with the software. In this phase, there are three mechanisms ensuring reproducible training:

1. The task description contains a detailed description of the tasks, how to solve them and explanations of the software used.
2. A demo video with a duration of circa five minutes guides the user through the study process, again explaining key points of the task description.
3. A training software is provided, in which the user can perform ten tasks which are structured the same as in the following measurement phase.

In all of those three mechanisms, the presented examples are not from the Shanghai MS, but taken from the Lu edition, to ensure proper methodological distinction between training and measurement phases.

During this phase, the study leader assists the participant in an online call with screen sharing, explaining the task description sheet, and answering their questions. The training phase's duration is approximately 15-20 minutes.

After this initial training phase, the user is expected to complete the second phase (measurement phase) on their own and without assistance by the study leader. In this phase, the participant is shown 719 or 720 image recognition tasks (the exact number depends on the task set assigned to the user) taken from the corresponding subset of the Shanghai MS.

To enter the measurement phase, the user opens a link provided in the task description sheet, and they are asked to enter the login credentials (which are unique to each user and printed on the description sheet) to start the measurement. Then, the user's annotations for each sample are recorded, together with a timestamp for each of the user's actions. In summary, only the individual sample annotations, the accuracy metrics (total accuracy, pitch accuracy, and secondary accuracy), and the time the user needed to complete all tasks (calculated as the sum of time differences between user actions, where each difference is capped at 5 minutes) are recorded.

The user can freely choose when they start the measurement phase, however they are asked to perform the measurement in a calm environment and to not interrupt it in order to keep the results comparable (e.g., after a long break the concentration might be better, thus positively influencing the accuracy) and the measurements traceable (since the measurement of time until study completion depends on timestamps of individual user actions).

Ethical considerations. Using the informed consent form (see Appendix E.2), the users are informed about the study's goals, the data that is collected from them, the absence of risks when participating in the study, their right to cancel participation in the study at any given time, their compensation, how long their data is stored, and other key points related to privacy and ethics.

	Accuracy (%)			CER (%)
	Pitch	Secondary	Total	Total
Average	87.2 ± 10.0	90.8 ± 5.0	84.1 ± 10.0	15.9 ± 10.0
Best	94.7	96.1	92.4	7.6

Table 6.6: Here, the accuracy values (for pitch classification, secondary classification, and total classification) for the user studies are presented. In the first row, the average result is shown, while the values for the user with the best total accuracy are contained in the second row.

As mentioned, during the measurement phase, only individual annotations, the accuracy and time metrics are collected. In addition, the participant's age, gender, the presence of normal or corrected visual acuity, and whether they have knowledge of a Chinese language or *suzipu* notation is recorded. These values are used to provide information in an aggregated manner (such as average age or gender distribution inside the sample), and to make sure that the participant fulfills the basic requirements for participating in the study. This information is under no circumstances linked to the data collected during the measurement phase. All data and the signed informed consent sheets are stored for ten years, after which they are permanently destroyed.

Since this user study serves the purpose of comparing the performance of humans with the performance of the OMR models developed in the previous sections, an indirect influence of the study on human decision-making processes cannot be excluded. However, the OMR models are designed to facilitate the annotation of music scores containing *suzipu* notation, and to guarantee a high quality of the produced annotations, human expertise is needed. Deskillling of experts is not expected, since the study of *suzipu* notation is currently a niche area of expertise, so the publication of digital toolboxes or AI models dealing with this kind of notation is more likely to lead to increased exposure and interest among the international research community.

The user study has been approved by the Ethics Committee of the Graz University of Technology (Reference Number EK-64/2025), and its approval is found in Appendix E.3.

Study results. All 15 participants have completed the study successfully. The average time needed for completion of the study is 76.3 ± 19.3 minutes. Accounting for the different numbers of presented samples, the average time to complete a whole edition is therefore 152.5 ± 38.6 minutes.

In addition to the time, different accuracy metrics (pitch accuracy, secondary accuracy, total accuracy, and total CER) have been collected in the study. The aggregated metrics and the metrics of the best performing participant are found in Table 6.6. The standard deviation is very high, and also the best user's CER is with 7.6% high compared to other optical recognition tasks.

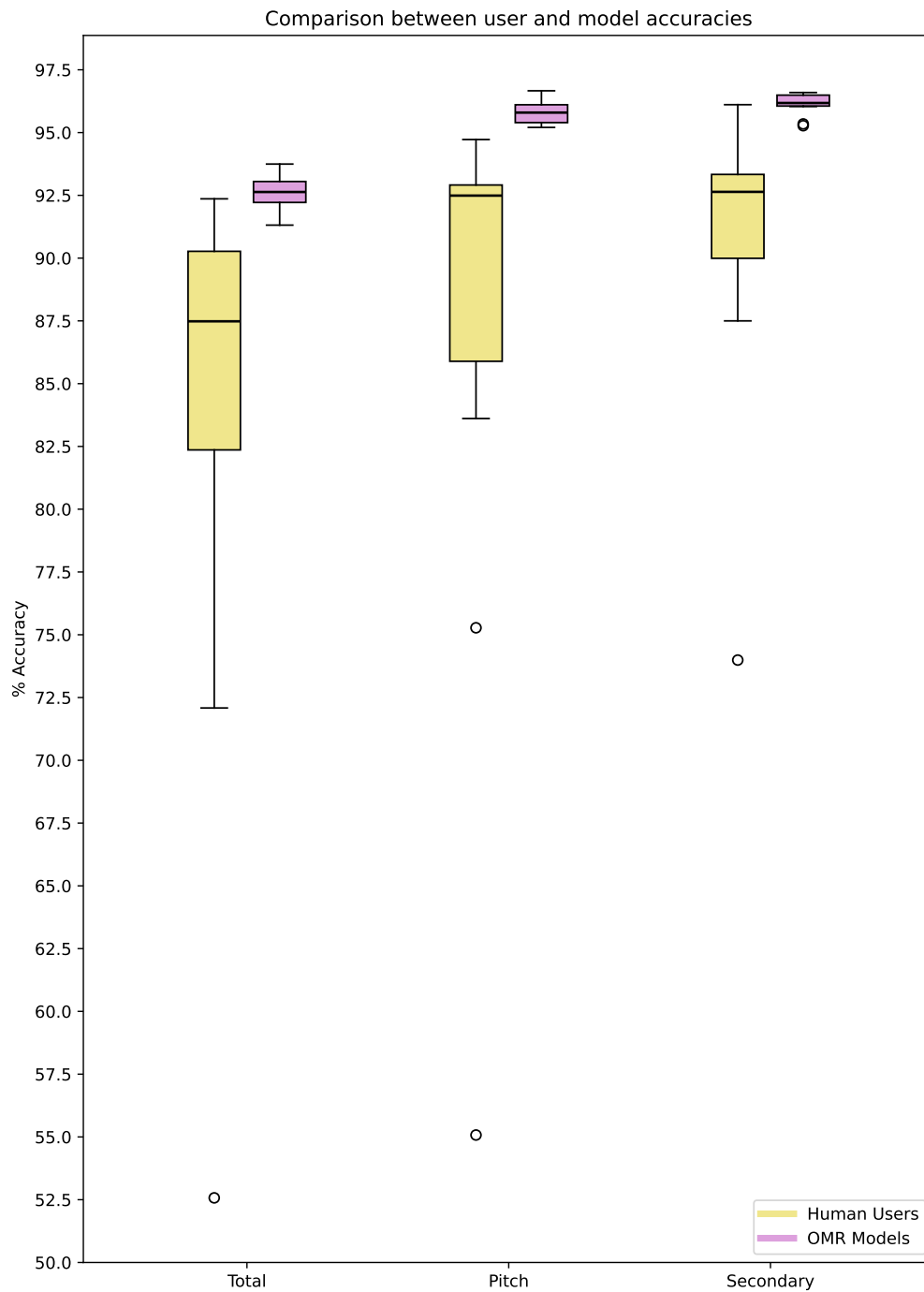


Figure 6.15: The accuracy boxplots (total accuracy, pitch accuracy, and secondary accuracy) clearly indicate that the OMR models' performance is better than the human users'. Note that there are 15 samples for the values associated with human users. Regarding the OMR models, there are ten samples for each pitch and secondary values, while for the total value there are 100 samples.

Comparing the users' accuracies with the OMR models' using a Wilcoxon rank-sum test ($\alpha = 0.01$) yields that the median OMR accuracy is significantly ($p = 1.9 \cdot 10^{-7} < \alpha$) larger than the humans'. This non-parametric test was chosen due to the small sample size and the stark negative skew in the human accuracy values. The boxplots of collected human values and a comparison with the values of the OMR classifier for total accuracy, pitch accuracy and secondary accuracy are found in Figure 6.15. Here, it can easily be seen that the OMR classifiers' lowest 25%-quantile is contained inside the humans' highest 25%-quantile.

6.8 Interpretation

Suzipu classification. For the *suzipu* classification task, which involves $11 \cdot 7 = 77$ classes, the aggregated CER is $6.6\% \pm 1.2\%$. This means that a user must correct only about every 20th annotation generated by the model. In comparison with the similar task of isolated character classification of Ethiopic script, which is based on a product space composition into $33 \cdot 7 = 231$ classes, the model by Belay et al. (2019) exhibits a CER of 5.03%, where the size of the total dataset is with 77994 instances roughly ten times larger than the number of *suzipu* instances in the KuiSCIMA dataset.

Testing on the Shanghai MS, the CER of 7.1% for the selected model is a clear improvement against our previous baseline model (Repolusk & Veas, 2024c) with 10.4%. Furthermore, this value is not only better than the average human-level performance (having a CER of $15.9\% \pm 10.0\%$), but also outperforms the best-performing human user that had a CER of 7.6%. This strongly indicates that the OMR models are reliable for the task of *suzipu* classification, despite the small dataset used for training.

The per-class F_1 scores for the best OMR model and the two best human users (spanning the whole Shanghai MS) in Figure 6.16 indicate that both the model and the humans are similar in that they both perform well on very frequent classes (e.g., the class *Che/None* 人). Vice versa, classes where humans (e.g., *Gong/Dadun* 𐌌) or the model (e.g., *Gong/Xiaozhu* 𐌌) have low performance are infrequent classes. The average per-class F_1 scores for the model (0.79 ± 0.26) and the human user (0.79 ± 0.27) show a similar performance on individual class level, even though the classes where F_1 is low are different.

Also regarding time needed for annotation, the *suzipu* model takes 2.04 ± 0.22 seconds, while it takes the average user 152.5 ± 38.6 minutes. Thus, the classifier's accuracy is higher and the time needed is significantly lower than the human users'.

Lülüpu classification. The *lülüpu* classification task involves 17 distinct classes, and the OMR models perform very well with an aggregated CER of $0.9\% \pm 0.9\%$. In other words, a human expert must on average correct every 100th model annotation. The Tesseract CER model's performance is with a CER of 51.6% not comparable with the CNN model. Also regarding timing, the *lülüpu* CNN is with 0.53 ± 0.07 seconds (500 evaluations) significantly faster than the Tesseract model with 7.9 ± 0.1 minutes.

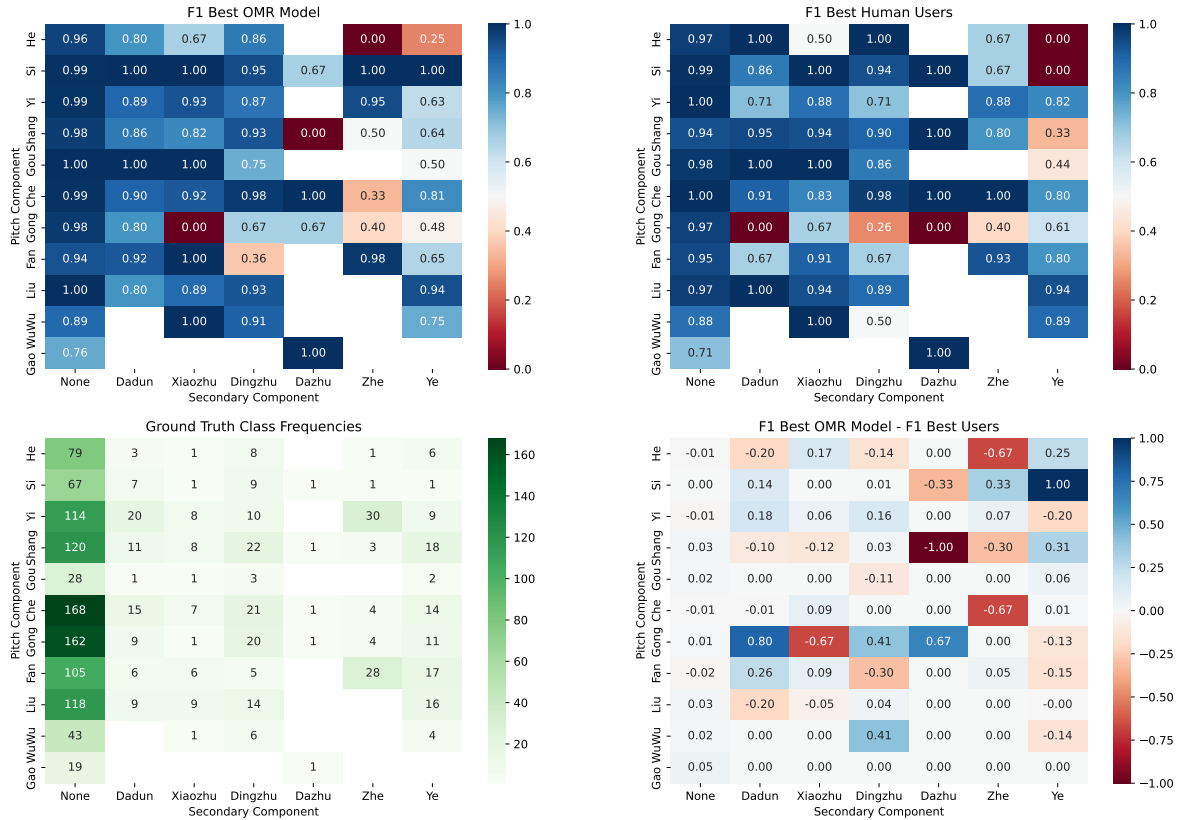


Figure 6.16: The heatmaps show the F_1 scores for the best OMR model (top left), the best two human users (top right), their difference (bottom right) and the absolute class frequency in the Shanghai MS (bottom left).

Both classification tasks. The test results on both the *suzipu* and *lülüpu* classifiers exhibit significantly lower accuracies whenever the Zhu edition is taken as a test set. This indicates that the distribution of the Zhu dataset’s image data might be considerably different from the other editions. From this and the results about variability by Martinez-Sevilla et al. (2023), one can conclude that all other editions do not introduce as much data variability each as the Zhu edition, since for them, high accuracy values are achieved despite dataset exclusion in training and vice versa.

Using temperature scaling, the calibration of the CNN models for *lülüpu* classification can be reduced significantly, while the *suzipu* classifiers are already well-calibrated without it. All ECE values (for 10 equidistant bins) are in the range of 0.0062 to 0.0162, thus constituting well-calibrated models.

Finally, the classifiers’ feature embeddings have been visualized using UMAP, showing a good separation between individual clusters that correspond well to ground-truth labels.

6.9 Summary

In this chapter, we have successfully used our KuiSCIMA dataset to create four different algorithms acting on image data. First, we developed raw image clustering for a quick overview of already annotated dataset instances. Then, we created and calibrated lightweight CNNs models for the OMR of *suzipu* and *lülüpu*.

These models were extensively evaluated: On the one hand using dataset splits where one full edition each is used for testing; and on the other hand, using an out-of-the-box Tesseract OCR model for the character-based *lülüpu* notation, and a newly created user study for estimating the *suzipu* model’s performance against a baseline of 15 naive human users. Using a feature encoding that is extracted from the OMR models, we create a UMAP-based clustering approach for additional verification that our models are feasible.

The interpretation at the end of this chapter illuminates model behavior and verifies that the developed approaches exhibit state-of-the-art performance.

We can now make use of the KuiSCIMA corpus in another way: By mathematically formalizing the music-theoretic rules of Chapter 3 and combining them with data-driven trends extracted from the dataset, we can tackle Research Question (RQ4) and create a system that generates stylistic music in the style of the 17 *suzipu* pieces in Chapter 7.